Next Generation Data Centers and Energy Management Systems

Lecture 9A: Machine Learning and Optimization Tools

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Berkeley
UNIVERSITY OF CALIFORNIA

Learning Plan

- Introduction to Data Center Systems and Thermodynamics
- Psychrometrics and Heat Transfer
- Performance Metrics and Standards
- Applied Computational Fluid Dynamics (CFD)
- Liquid Cooling & Advanced Heat Transfer
- Chip Cooling: Heat Conduction and Thermal Interface Materials
- Power Delivery and Infrastructure (Energy sources: Geothermal, Solar, Nuclear, etc.)
- Overview of Simulation Field; the Concept of a Digital Twin
- **Machine Learning and Optimization Tools**
- Modeling and Simulation of an Energy Management System
- Control Systems and Optimization
- Spacecraft Thermal Control
- Final Project Work, Peer Feedback, and Mentor Check-in
- Symposium

Learning Objectives

- Develop and apply a genetic algorithm to optimizing rack layout in a data center

Typical layout of a data center facility

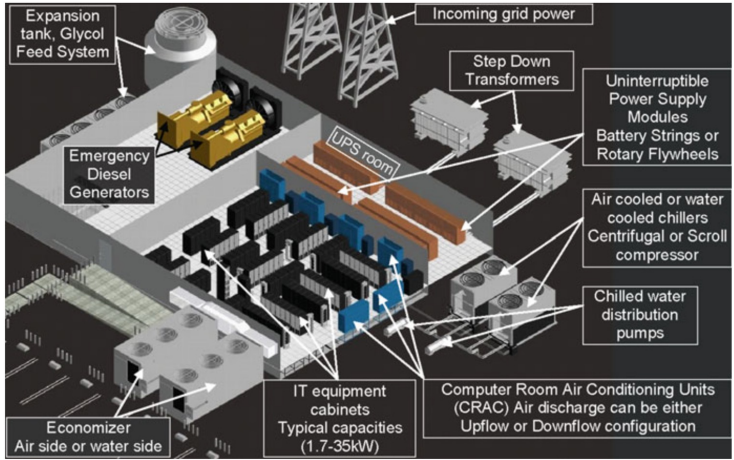


Figure 1: (Figure 1.1 in Ref. [1]).

Seminar 6

- Surya Swamy and Aniket Soneji, Hammerhead AI
- Title: Turning Stranded Power into AI Inference-ready Capacity
- Date: October 30, 2026
- Time: 10:00 (Berkeley Time)
- Location: EH 3110 (Seminar Room)

Seminar 7

- Sherman Ikemoto, Cadence
- Title: Dynamic Infrastructure: Digital Twin Architecture for Next-Generation AI Factories
- Date: November 5, 2026
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Seminar 8

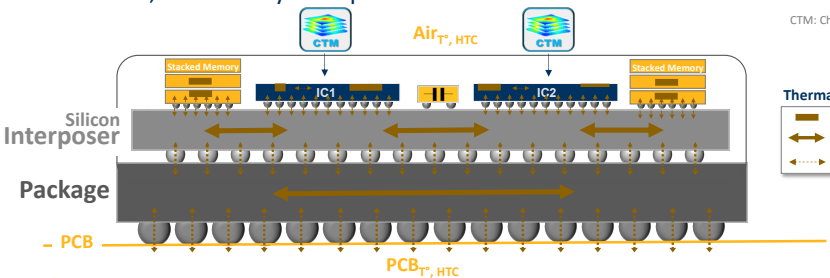
- Alex Lotz, AWS
- Title: TBD
- Date: November 5, 2026
- Time: 15:30 (Berkeley Time)
- Location: GSPP 150

Chip Thermal Modeling for 3DICs

- Predict the thermal hotspot of the die across multiple on-die routing layers
- Hotspot locations can guide on-die sensor placement and thermal management
- Chip/chiplet level static and transient temperature response
- RedHawk-SC Electrothermal is our chip-level thermal sign-off solution, certified by multiple foundries



CTM: Chip Thermal Model



Thermal Conductive Paths

sources	✓ IC Power Density
intrinsic	✓ Metal Densities
TSV, Bumps	✓ Materials
	✓ Boundary Conditions

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Digital Twin Introduction

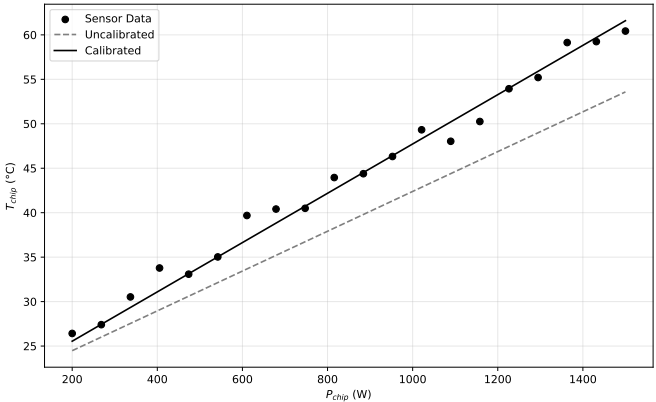


Figure 2: Calibrating a digital twin.

Reduced Order Modeling

References I

[1] Yogendra Joshi and Pramod Kumar, editors.
Energy Efficient Thermal Management of Data Centers.
Springer US, 2012.